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### VEHICLE LID STRUCTURE

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#### Abstract

A vehicle lid structure includes: a main body member that is able to block an opening formed on an exterior surface visible from the outside; and a base material which supports the main body member and is swingable about a shaft member located behind the exterior surface. The base material has an edge cover portion which covers an edge of the main body member from the rear side. When viewed from a direction along the exterior surface based on a closed state, a surface of the edge cover portion is inclined to approach the opening as it goes from a center of the opening toward an edge of the opening.

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#### Background/Summary

## FIELD OF THE INVENTION

[0001] The present invention relates to a vehicle lid structure.

## BACKGROUND

[0002] A fuel filler port and a charging port of a vehicle are often designed to be opened and closed using a lid device called a fuel lid. A conventional technique related to such a lid device is disclosed in Japanese Patent Application Laid-Open No. 2019-98807. The lid device is configured to open and close an opening in a vehicle body by using a swingable lid.

[0003] Incidentally, in a lid device in which a design surface is exposed to the outside, it is anticipated that a non-occupant may improperly pry open the lid using a tool or the like. There is room for improvement in terms of prying the lid open.

[0004] An object of the present disclosure is to provide a vehicle lid structure having a lid that is difficult to pry open.

## SUMMARY

[0005] According to the present disclosure, there is provided a vehicle lid structure including: a main body member that is able to block an opening formed on an exterior surface visible from the outside; and a base material which supports the main body member and is swingable about a shaft member located behind the exterior surface, wherein the base material has an edge cover portion which covers an edge of the main body member from the rear side, and wherein when viewed from a direction along the exterior surface based on a closed state, a surface of the edge cover portion is inclined to approach the opening as it goes from a center of the opening toward an edge of the opening.

[0006] According to the present disclosure, it is possible to provide a vehicle lid structure having a lid that is difficult to pry open.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a front view of a lid device that adopts a vehicle lid structure according to an embodiment;

[0008] FIG. 2 is a cross-sectional view taken along line 2-2 of FIG. 1;

[0009] FIG. 3 is an enlarged perspective view of an edge cover portion shown in FIG. 2; and

[0010] FIG. 4 is an enlarged view of part 4 of FIG. 2.

### DETAILED DESCRIPTION

[0011] An embodiment of the invention will be described below with reference to the accompanying drawings. The embodiment shown in the accompanying drawings are merely examples of the invention, and the invention is not limited to this embodiment.

#### Embodiment

[0012] A description will be made with reference to FIG. 1. FIG. 1 shows a vehicle lid device 20 (hereinafter, referred to as “lid device 20”) using a vehicle lid structure. The lid device 20 is, for example, a fuel lid that opens and closes a fuel filler port or a charging port. The lid device 20 opens and closes an opening 11 provided in an exterior member 10 provided on the vehicle body.

[0013] The exterior member 10 is a component made of resin. The exterior member 10 is, for example, a front grille provided at the front of the vehicle. The surface of the exterior member 10 facing the exterior of the vehicle is referred to as an “exterior surface 12” that is visible from the outside. The exterior member 10 has the opening 11 that is a substantially rectangular hole.

[0014] A description will be made with reference to FIG. 2. A plurality of bosses 14 and 15 for fixing the lid device 20 are formed on an exterior rear surface 13 which is the rear surface of the exterior surface 12. The lid device 20 is fixed to the bosses 14 and 15 by screw fastening members

**17** and **18** formed by screws.

[0015] The lid device **20** includes a support member **30** that is fixed to the rear side of the exterior member **10**, a shaft member **22** which is a shaft supported by the support member **30**, and a lid **40** which is supported by the shaft member **22** and is able to open and close the opening **11**.

[0016] A description will be made with reference to FIG. **1**. The edges of the substantially rectangular opening **11** correspond to the respective sides, and are referred to as a first edge portion **11a**, a second edge portion **11b**, a third edge portion **11c**, and a fourth edge portion **11d**, respectively, from the edge on the left side in the figure.

[0017] A description will be made with reference to FIG. **1** and FIG. **2**. The first edge portion **11a** and the third edge portion **11c** extend substantially in parallel in the extension direction of the shaft member **22**, and the second edge portion **11b** and the fourth edge portion **11d** extend substantially perpendicularly to the extension direction of the shaft member **22**. The first edge portion **11a** is located at a position closer to the shaft member **22** than the third edge portion **11c**. The third edge portion **11c** is located at a position farther from the shaft member **22** than the first edge portion **11a**.

[0018] A description will be made with reference to FIG. **2**. In the bosses **14** and **15**, the boss **14** formed in a position closer to the first edge portion **11a** is referred to as the first boss **14**, and the boss **15** formed in a position closer to the third edge portion **11c** is referred to as the second boss **15**.

[0019] The screw fastening members **17** and **18** may be any type of screw, such as a machine screw, a tapping screw, a cap bolt, or a bolt. In the screw fastening members **17** and **18**, the screw fastening member **17** fastened to the first boss **14** is referred to as the first fastening member **17**, and the screw fastening member **18** fastened to the second boss **15** is referred to as the second fastening member **18**.

[0020] The support member **30** is configured as an integrally molded article. The support member **30** is a substantially box-shaped member with an opening at the portion where the lid **40** is attached. The support member **30** has a first fixing portion **31** which is fixed to the first boss **14**, a second fixing portion **32** which is fixed to the second boss **15**, a first side wall **33** which is connected to the end portion of the first fixing portion **31** and extends in a direction away from the exterior member **10**, a second side wall **34** which is connected to the end portion of the second fixing portion **32** and extends in a direction away from the exterior member **10**, a support member contact portion **35** which extends along the rear surface of the exterior member **10** from an end portion (one end) of the first side wall **33** closer to the exterior member **10** and comes into contact with the exterior rear surface **13** of the exterior member **10**, a rear wall **36** which connects an end portion (the other end) of the first side wall **33** farther from the exterior member **10** to an end portion (the other end) of the second side wall **34** farther from the exterior member **10**, a bottom portion **37** which closes the lower end, and a ceiling portion (not shown) which faces the bottom portion **37** and closes the upper end. The support member contact portion **35** extends along the exterior rear surface **13** from the first side wall **33** toward the edge of the opening **11**.

[0021] The lid **40** has a base material **50** which is supported by the shaft member **22**, a main body member **42** which is supported by the base material **50** and is exposed to the outside of the vehicle, and a held portion **43** which extends from the tip of the base material **50** toward the support member **30** and is held by a locking device (not shown) in a fully closed state.

[0022] The base material **50** is configured as a single resinous molded product. The base material **50** has a lid contact portion **51** which extends in the radial direction from the shaft member **22** and is able to contact the support member **30** in a fully open state, a connection portion **52** which connects the tip of the lid contact portion **51** to the end portion of the main body member **42** on the side of the first edge portion **11a**, a space forming portion **53** which is provided along the rear surface of the main body member **42** and forms a space together with the main body member **42**, and an edge cover portion **54** which is formed at the tip of the end portion of the space forming portion **53** and covers an edge **42e** from the rear side of the main body member **42**.

[0023] The lid contact portion **51** is provided with a lid contact portion rib **51a** which comes into

contact with the support member **30** in a fully open state and increases the strength of the lid contact portion **51**.

[0024] The connection portion **52** is formed in a curved shape not to come into contact with the first edge portion **11a** when displacing from the fully closed state to the fully open state.

[0025] A description will be made with reference to FIG. **3** and FIG. **4**. The edge cover portion **54** has a plate-shaped portion **54a** which is formed in a tapered shape to gradually approach the edge **42e** of the main body member **42** from the tip of the space forming portion **53**, an edge cover base portion **54b** which extends from the tip of the plate-shaped portion **54a** toward a rear surface **42f** of the main body member **42**, a positioning protrusion **54c** which protrudes from the tip of the edge cover base portion **54b** and determines the relative positions of the main body member **42** and the base material **50**, an edge cover portion rib **54d** (rib **54d**) which extends from a base end **54f** (a boundary between the space forming portion **53** and the plate-shaped portion **54a**) of the plate-shaped portion **54a** toward the rear surface **42f** of the main body member **42**, and an edge cover connection portion **54e** (connection portion **54e**) which connects the plate-shaped portion **54a**, the edge cover base portion **54b**, and the edge cover portion rib **54d**.

[0026] Furthermore, when the positioning protrusion **54c** comes into contact with the rear surface **42f**, the edge cover base portion **54b** is separated from the rear surface **42f**. The edge cover portion **54** may not be provided with the positioning protrusion **54c**. When the positioning protrusion **54c** is not provided, the edge cover base portion **54b** comes into contact with the rear surface **42f**. Further, the edge cover portion rib **54d** does not necessarily come into contact with the rear surface **42f** regardless of the presence or absence of the positioning protrusion **54c**.

[0027] A description will be made with reference to FIG. **1** and FIG. **2**. The main body member **42** is configured as an integrally molded article. A lid design surface **42a** which is a surface exposed to the outside of the vehicle in the main body member **42** is substantially continuous to the exterior surface **12**. The lid design surface **42a** has a substantially rectangular shape along the opening **11**.

[0028] A part of the lid design surface **42a** may be formed as a recess **42b** that is slightly recessed compared to other parts. Accordingly, an operator can easily recognize the part to be operated by the operator.

[0029] A main body protrusion **42c** protrudes from the edge **42e** of the main body member **42** along the side surface of the edge cover portion **54**.

[0030] An opening protrusion **11e** which protrudes toward the inside of the vehicle is integrally formed at the edge of the opening **11**.

[0031] The vehicle lid structure adopted in the lid device **20** (lid **40**) described above will be summarized below.

[0032] A description will be made with reference to FIG. **2**. Firstly, the vehicle lid structure includes the main body member **42** that is able to block the opening **11** formed on the exterior surface **12** which is visible from the outside and the base material **50** which supports the main body member **42** and is swingable about the shaft member **22** located behind the exterior surface **12**. The base material **50** has the edge cover portion **54** which covers the edge of the main body member **42** from the rear side. When viewed from a direction along the exterior surface **12** based on the closed state, the surface of the edge cover portion **54** is inclined to approach the opening **11** as it goes from the center of the opening **11** toward the edge of the opening **11**.

[0033] A description will be made with reference to FIG. **4**. Since the surface of the edge cover portion **54** is inclined, the tool T is less likely to get caught, and prying open can be suppressed. Further, since the edge cover portion **54** is provided along the edge **42e** of the main body member **42**, the load from the tool T is received by two members, the base material **50** and the main body member **42** when the tool T is inserted for an opening operation. The load can be dispersed more effectively than when the load is received by the main body member **42** alone, and hence the main body member **42** is less likely to be broken. Since the main body member **42** located on the outer side is less likely to be broken, prying open can be further suppressed.

[0034] Additionally, since the edge cover portion **54** is provided, the appearance when the **40** is open is improved. In particular, since the surface is inclined, the main body member **42** and the base material **50** are more integrated in appearance, and hence the appearance is further improved. [0035] Secondly, in a first vehicle lid structure, the edge cover portion **54** has the plate-shaped portion **54a** including the surface of the edge cover portion **54** and the edge cover portion rib **54d** (rib **54d**) erected on a rear surface of the plate-shaped portion **54a**. The edge cover portion **54** is composed of the plate-shaped portion **54a** and the edge cover portion rib **54d**. The lid structure can be made lightweight and highly rigid.

[0036] In the embodiment, the edge cover portion rib **54d** extends from the base end **54f** of the plate-shaped portion **54a** toward the rear surface **42f** of the main body member **42**. The base end **54f** of the plate-shaped portion **54a** is the farthest the rear surface **42f** of the tapered plate-shaped portion **54a**. Therefore, the edge cover portion rib **54d** can be set higher, and the edge cover portion **54** can be reinforced more reliably.

[0037] Furthermore, the edge cover portion rib **54d** is located between the held portion **43** of the locking mechanism and the main body member **42**. Accordingly, it is possible to suppress the loss of load when locking the lid **40** and to smoothly lock the lid **40**.

[0038] Although the vehicle lid structure according to the invention has been described as being applied to a fuel lid, the vehicle lid structure can also be applied to other devices. Further, the shape of the main body member **42** is not limited to a substantially rectangular shape, but may be any shape such as an ellipse or a circle.

[0039] In other words, the invention is not limited to the examples as long as the functions and effects of the invention are achieved.

## Claims

1. A vehicle lid structure comprising: a main body member that is able to block an opening formed on an exterior surface visible from the outside; and a base material which supports the main body member and is swingable about a shaft member located behind the exterior surface, wherein the base material has an edge cover portion which covers an edge of the main body member from the rear side, and wherein when viewed from a direction along the exterior surface based on a closed state, a surface of the edge cover portion is inclined to approach the opening as it goes from a center of the opening toward an edge of the opening.
  2. The vehicle lid structure according to claim 1, wherein the edge cover portion has a plate-shaped portion which includes the surface of the edge cover portion and a rib which is erected on a rear surface of the plate-shaped portion.
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